Movie Recommendation System with Sentiment Analysis

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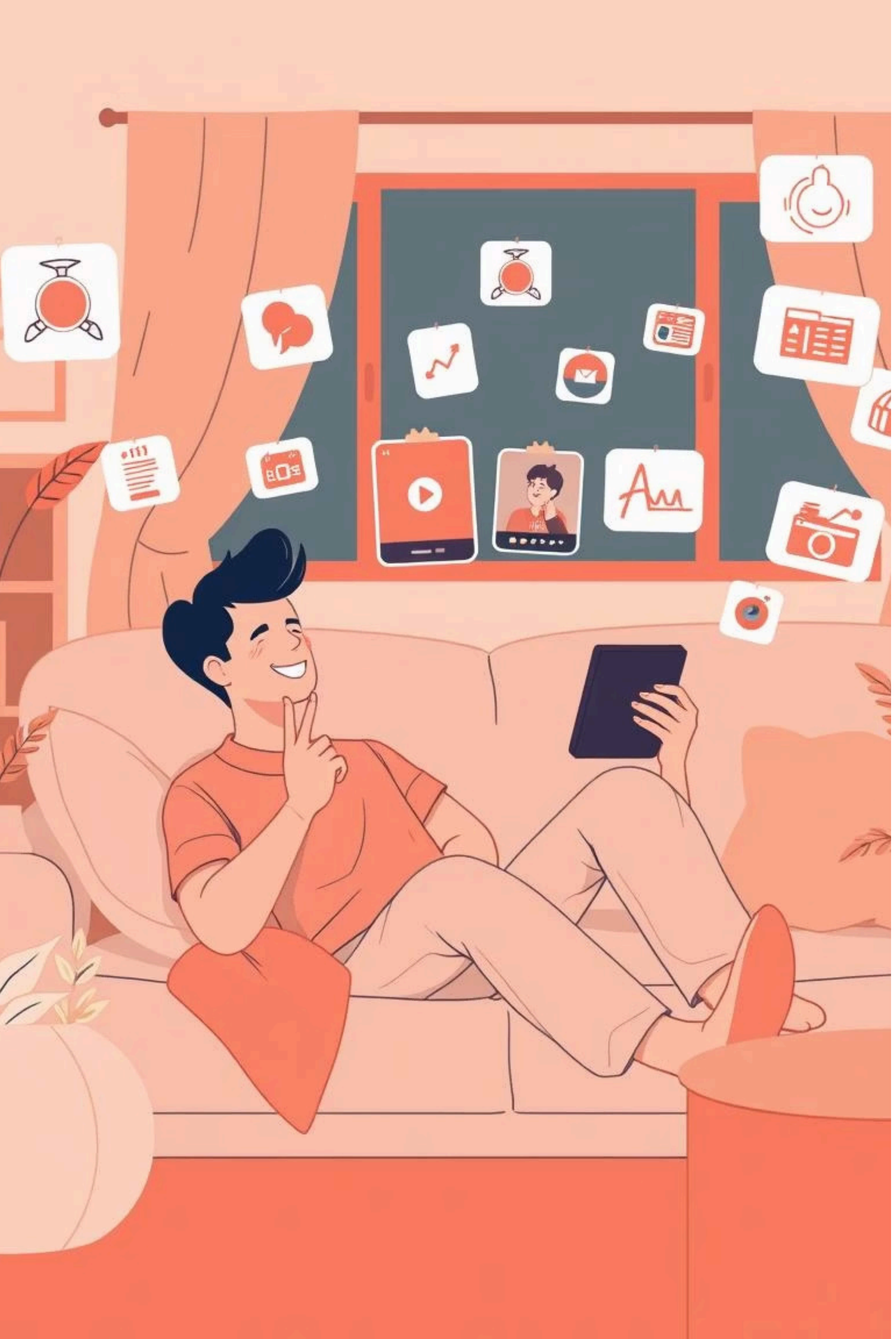


Introduction

Project Overview

A web-based application to enhance the movie discovery experience for users. It provides personalized recommendations and integrates sentiment analysis of user reviews.

Purpose

To offer a comprehensive platform that combines content-based recommendations with real-world public opinion.

Problem Statement

Information Overload

Users are overwhelmed by the sheer number of movies available.

Lack of Contextual Recommendations

Existing systems may not provide insights into the general sentiment surrounding a movie.

Scattered Information

Users often have to consult multiple sources to find movie details and reviews.

Project Objectives

1. **Develop a user-friendly web interface**

   for searching movies and receiving recommendations.

2. **Provide content-based movie recommendations**

   based on movie features like genre, cast, and plot.

3. **Integrate with the TMDb API**

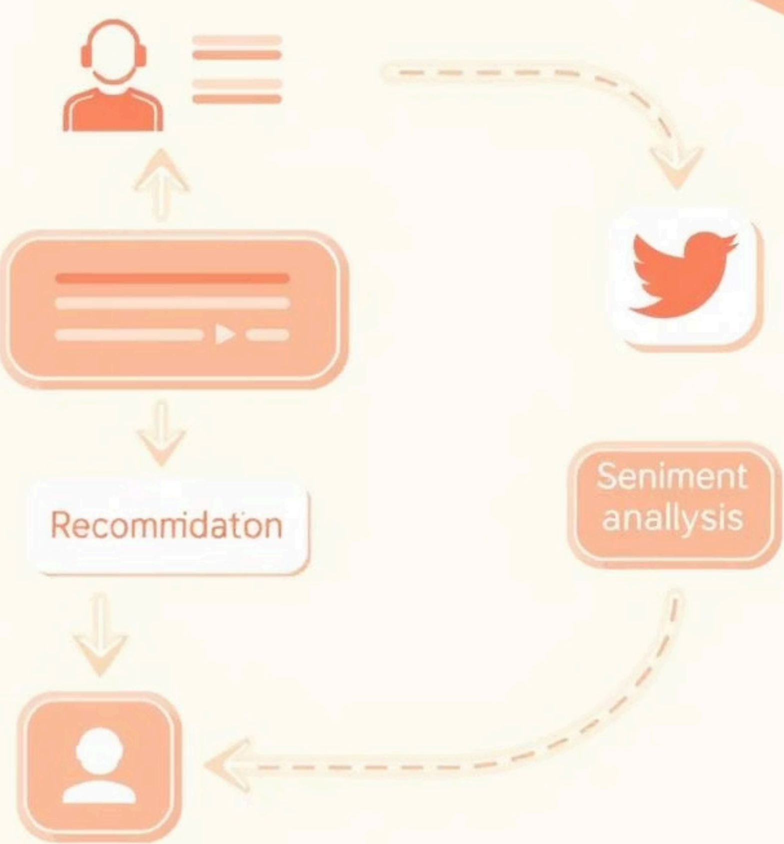
   to collect user reviews and provide detailed movie information.

4. **Classify fetched TMDb reviews**

   as positive or negative using sentiment analysis.

5. **Utilize AJAX**

   for asynchronous communication to ensure a smooth, interactive user experience.



Proposed Solution & Methodology

Interactive Search

An autocomplete feature for easy movie title input.

Recommendation Engine

Uses cosine similarity on vectorized movie features to suggest similar movies.

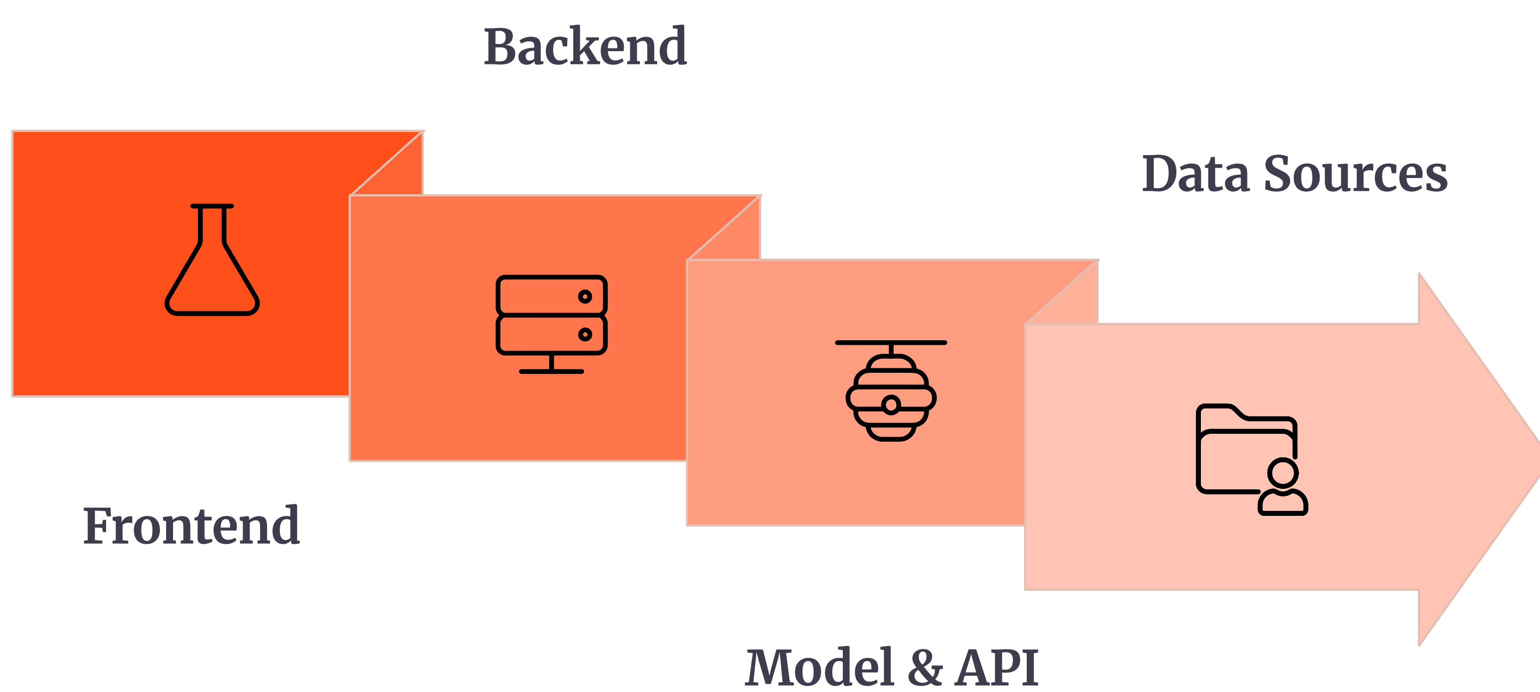
Sentiment Analysis

A pre-trained Multinomial Naive Bayes model is used to classify fetched reviews as positive or negative.

Data Flow

Reviews are pre-processed and vectorized using a TF-IDF vectorizer before sentiment prediction.

System Architecture



Frontend

Built with HTML, CSS, and JavaScript, using jQuery to simplify interactions and Bootstrap for design.

Backend

Powered by Python Flask, which handles user requests, fetches data, and performs the sentiment analysis.

Data Sources

TMDb API: Used to fetch real-time user reviews, movie details, and cast information.

Local Data: A pre-processed dataset (main_data.csv) is used for generating recommendations.

Tools and Technologies

Languages

Python (Backend), HTML, CSS, JavaScript (Frontend).

Frameworks & Libraries

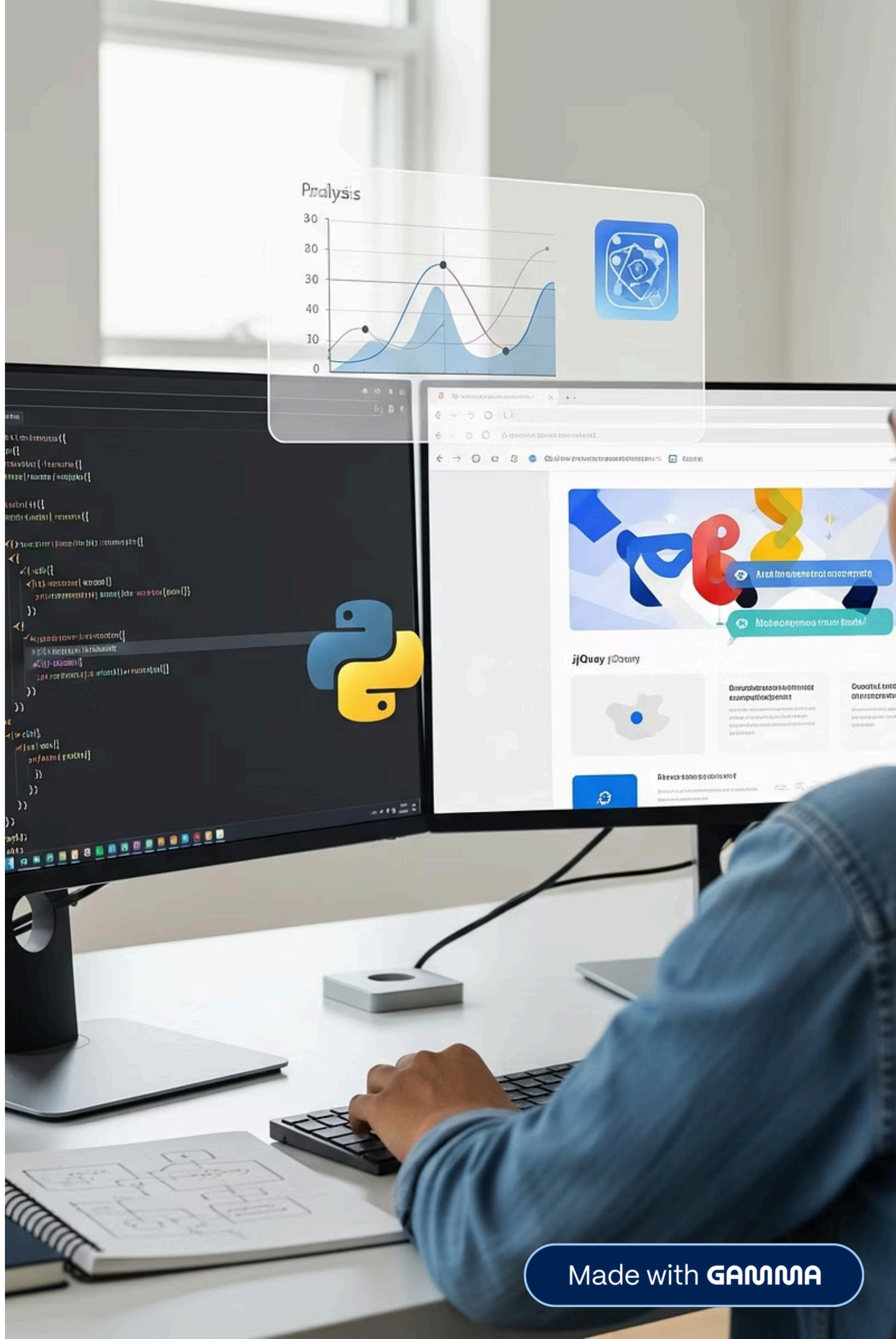
Flask, jQuery, scikit-learn, Pandas, NLTK, Requests.

APIs

TMDb API.

Models

Pre-trained Multinomial Naive Bayes classifier and TF-IDF vectorizer.



Implementation Details

1

Core Modules

The project structure includes the main Flask application (main.py), a data preparation script (fetch_and_preprocess_data.py), and frontend JavaScript logic (recommend.js).

2

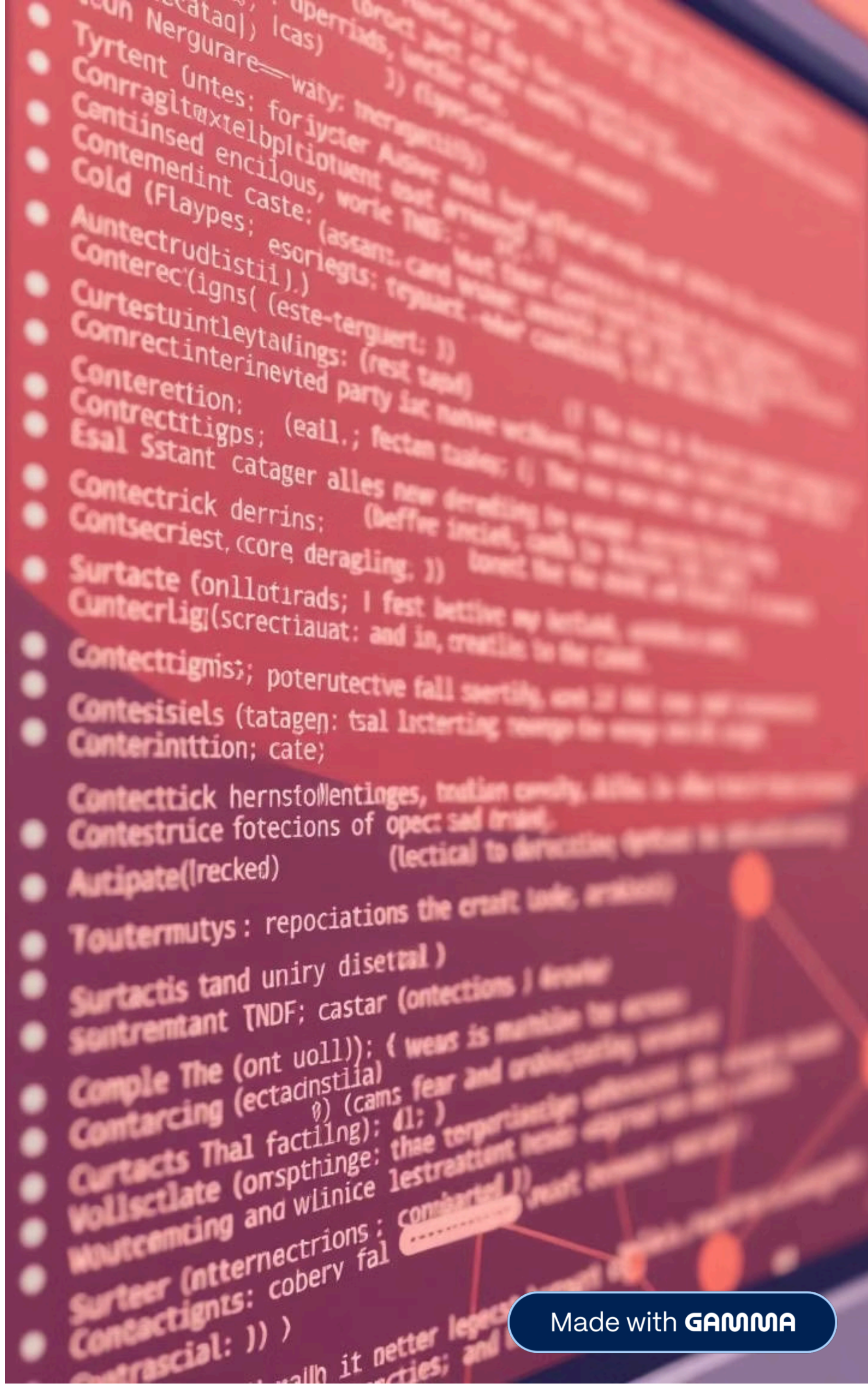
Dynamic Updates

AJAX is extensively used for seamless user interaction, preventing full-page reloads. This makes the user interface feel more responsive and fluid.

3

Model Persistence

The trained machine learning models are saved as .pkl files and loaded once at application startup to reduce latency.



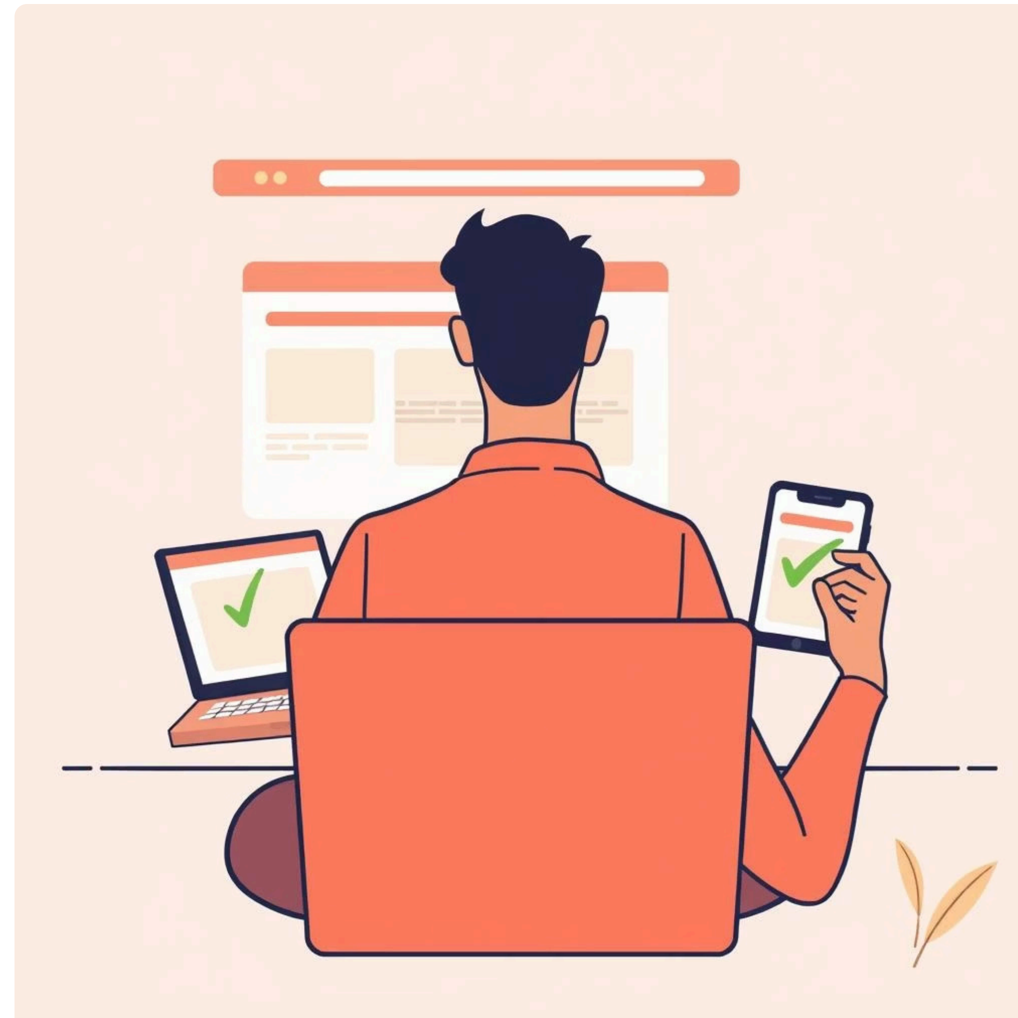
Testing and Results

Testing Methodology

Includes UI testing, recommendation engine testing, and sentiment analysis module testing.

UI Testing

Verified a user-friendly and responsive design across different screen resolutions.



Results

The system effectively generates accurate content-based recommendations and reliably classifies review sentiments.

Conclusion

The project successfully integrates a content-based recommendation engine with a real-time sentiment analysis module.

It effectively addresses the challenges of information overload and the need for contextual recommendations in the movie discovery process.

The project successfully combines data acquisition, machine learning, and web development to create a practical solution.

Future Scope

Implement collaborative filtering to provide recommendations based on user-to-user interactions.

Develop persistent user profiles and history to offer more personalized, long-term learning.

Explore advanced Natural Language Processing (NLP) techniques beyond the established models used in the project.